

CPE 663 Special Topic III: Deep Learning

Lab 5: Training a Neural Network

In this lab, we will implement a backpropagation algorithm to learn a simple deep forward network for classification task.

The following are the network architectural specification:

- Connection type: Fully connected
- Number of input units: As given in the dataset
- Number of hidden layers: 1
- Number of units in hidden layer: 6
- Activation function of hidden layer: ReLU
- Number of output unit: 1
- Activation function of output layer: Sigmoid

Tasks

1. Draw the diagram showing this deep forward network.
2. Write the equation of each units in the network (in hidden and output layer).
3. Write the error function for this training process.
4. Derive the gradient of weights in output layer.
5. Derive the gradient of weights in hidden layer.
6. Split the `cardio` dataset into training and test set. Use 70:30 proportion.
7. Implement standard gradient descent using the `training` set and plot the loss history across iterations. Then, test your trained network on `test` set.
8. Choose and implement the following neural network training techniques into your network:
 - **Category I** - Choose and implement `one` of the following :
 - Add a second hidden layer and updating the backpropagation chain rule accordingly.
 - Use an advanced optimizer (e.g., Nesterov, Adam, Adagrad, NAdam, or RMSProp) for weight updates.
 - Implement Batch Normalization on the hidden layer.
 - **Category II** - Choose and implement `two` of the following :
 - Apply L1 or L2 regularization during weight updates.
 - Implement a random noise injection in the learning process.
 - Implement a dropout mechanism to the learning process.
9. Compare the training process of your baseline network (from 7.) against the modified version (from 8.). Analyze and discuss any observed differences in the convergence rate, the final output weights, and the overall robustness when evaluated on the `test` dataset.

What to submit?

1. A short report (no more than 6 pages) which includes:
 - `.sbatch` configuration (partition, number of CPU, RAM, etc.).
 - A screenshot that shows Slurm job submission and result for each question.
 - Discussion on the result of the experiment (if any).
 2. A `.zip` file or link to Github/Gitlab repository contains:
 - Source code for each question.
 - `.sbatch` file.
 - Configuration file (if any).
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